Soudip Kundu

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Summary

I am currently completing a BS–MS Dual Degree Program in Physics at IISER Kolkata, with a focus on condensed matter physics. Through internships, workshops, and term papers, I have developed research experience across various topics in the field. I aim to further deepen my understanding of condensed matter systems and contribute to the study of strongly correlated quantum materials through my PhD research. My future goal would be to use this understanding to study other quantum phase transitions.

Education

2021 - 2026	BS-MS Degree at **Indian Institute of Science Education and Research Kolkata** (currently in 5th year)Transcript	(GPA: 8.38/10)
2020	Class 12th CBSE Board	(93.6/100)
2018	Class 10th CBSE Board	(91.6/100)

Research Experience

Current MS thesis(supervisor : Prof Siddhartha Lal [IISER Kolkata]) *5th year 2025-26*

- Aim is to understand the breakdown of the Fermi liquid entering into the pseudo gap phase and finally transition into the Mott insulator by solving a lattice embedded impurity model using unitary renormalization group(URG) at finite temperature
- URG is a technique developed by my professor and some of his PhD students. They have already studied this formation of pseudogap at zero temperature in one of their paper.Mott Criticality as the Confinement Transition of a Pseudogap-Mott Metal
- My role is to build a framework using the URG technique to study this transition at finite temperature.First I am learning this URG technique and understanding the Kondo model, like formation of singlet and flattening of the (spectral function)Kondo resonance at high temperature, and various thermodynamic properties.
- Next step will be to solve the single impurity Anderson model(siam) and then the ultimate goal is to put everything together and understand the transition from Fermi liquid to Mott insulator. What makes this URG technique better than others is that it is non-perturbative.

Summer Internship *2025*

I conducted an in-depth study of Landau–Fermi liquid theory and the Hatsugai–Kohmoto model, an exactly solvable model serving as a theoretical playground for understanding Mottness and non-Fermi-liquid physics. I also explored various aspects of quantum many-body physics, which formed an important foundation for my MS thesis work.

Summer Intership(Visiting student at HRI) *4th year 2024*

- Did my internship under Prof Sayan Choudhury, Reader at Harish Chandra Research Institute(HRI), on "**Time Crystal in periodically driven classical spin chain**"
- We considered a periodically driven Floquet system, a time-dependent Hamiltonian with discrete time symmetry. The breaking of the symmetry led to the novel state of matter DTC(Discrete Time Crystal), which is predicted by Prof Frank Wilczek.

- Prof Sayan Choudhury, in one of his papers, showed that if we change the periodicity of the Hamiltonian, then the formation of the stable time crystal is periodic, ie, for certain values of T= Time period of Hamiltonian, we see a stable DTC, but this is when we work with quantum spin(Pauli matrices)
- My job was to see whether we observe similar behaviour with the classical spin chain(Classical Ising Chain), which is periodically driven in time. And interestingly, we have found such periodic behavior here too, which can be seen in the phase diagram that we have got in the following report.Report

Summer Internship *3rd year 2023*

I have done an internship in Quantum Information and Computation and have also participated in quantum computing workshops and programs held by IBM and QWorld.Certificate

Term Papers *Click on the link to see the reports*

BCS to BEC crossover , Quantum Hall Effect , Determination of 1D band structure by transfer matrix method , Differential Form in Electromagnetism , Optical Fano Resonances

Professional Skill

Programing Skills

Proficient in Python for numerical simulations and data analysis; working knowledge of C++ and Julia.

Presentation Skills

Apart from my research experience, I have also taken time to develop particular skills that will prove invaluable in an academic career. I have given a presentation on my summer internship "Time crystal" in a student colloquium. The talk was well-received and gave me confidence in public speaking

Leadership Skills

I have organized IBM Quantum Fall Fest, where I gave lectures on the basics of quantum computing and error correction. Around 200 students in our institute participated in that event. It taught me to organize large-scale events and work with a group.

Teaching Assistantship *Jan 2- Present(2026)*

Basic Quantum Mechanics Course PH2201, Department of Physics, IISER Kolkata.

References

Prof Siddhartha Lal(IISER Kolkata)

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Group webpage:- EPQM

Sayan Choudhury(Reader at Harish-Chandra Research Institute(HRI))

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Prof Bhavtosh Bansal(IISER Kolkata)

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